> PRIMUS · 20 NOVEMBER 2026 · 150 TEAMS · ONE NIGHT

Organiser guide

Set up, run and close the PRIMUS night:
the control panel, Event timing and the AI judge.

For organisers, the fairness officer and the tech desk · confidential

Build round · Rescue round · Invest round · AI judges · live market

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1. The night at a glance

PRIMUS runs for one night, for about 600 participants in 150 teams. Every team plays three rounds (Build, Rescue, Invest) worth up to 100 points each. AI judges score the written work, and the scores move the share prices of a live market that all teams trade in. This guide is for the people who run it: organisers, the fairness officer, the tech on call, and whoever sits at the projector.

> **GOOD TO KNOW**
> **You control the clock.** Nothing starts until an organiser clicks **Start the night**, and nothing ends until an organiser clicks **End the night**. In between, every phase has a duration you can change, and each phase's deadlines follow from when it actually started. The times in this guide are the planned night.

Phases and what happens in each

Planned	Phase (default length)	What happens	Who acts
before	Setup	Content, logins, the lottery commitment. The clock does not run.	Organisers
20:00	Check-in (45 min)	Teams sign in with their cards.	Registration desk
20:45	Briefing (30 min)	Rules explained. **The lottery must be drawn before Build can start.**	Event lead
21:15	Round 1: Build (105 min)	Problem cards dealt; PRDs written. Twist 60 min in. PRD form locks at the end.	Teams; twist automatic
23:00	Judging the PRDs (30 min)	Three AI readings per PRD; scores sealed.	AI desk
23:30	PRD reveal (15 min)	Starting prices and the top 5 PRDs. Waits until every PRD score is sealed.	Automatic
23:45	The crisis (30 min)	Crisis cards, prices −15%, rescue circle, seed published; Q&A board.	Automatic
00:15	Round 2: Rescue (90 min)	Crisis solutions written; rumours. Form locks at the end.	Teams; rumours by organisers
01:45	Judging the solutions (30 min)	Three AI readings per solution; scores kept secret.	AI desk
02:15	Invest, blind (rounds 1–2)	Solutions published without scores; trading. Waits until every solution score is sealed.	Automatic
02:45	Invest (rounds 3–9)	The score reveal moves prices; trading. Leaderboard freezes 45 min in.	Exchange desk watches
04:30	Closing bell (45 min)	Closing prices, points, awards, collusion flags.	Fairness officer
05:15	Awards (untimed)	Results revealed one by one. Waits until every flag is decided.	Event lead ends the night

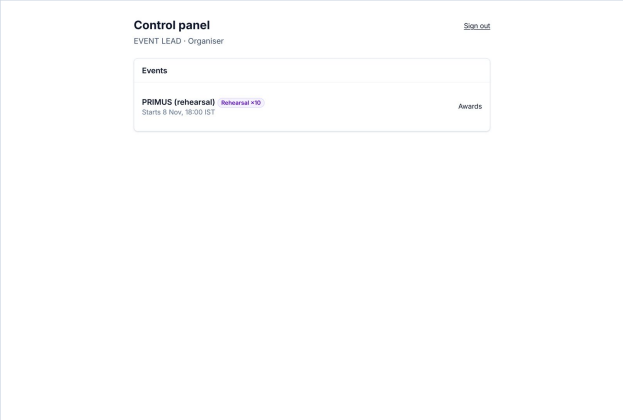
Who does what on the night

Role	Where	Job
Event lead (organiser)	Laptop 1, **Timing**	Starts the night, ends phases, runs the lottery on stage, pauses, extends, ends the night, talks to the room.

Role	Where	Job
AI desk (organiser)	Laptops 2 and 3, **Judge**	Keeps “Run the judge here” on during judging; watches every score seal; handles re-runs.
Exchange desk (organiser)	Laptop 4, **Rounds** and **Ledger**	Watches rounds open and clear; approves corrections and appeals (the second organiser).
Fairness officer (faculty)	Own laptop, **Fairness**	Reviews every collusion flag at the closing bell; clears or disqualifies with a reason.
Tech on call	**Health**, Supabase and Render dashboards	Connections, errors, the venue network.
Registration desk	Printed login cards	One card per team; reprints lost cards.
Projector	A laptop signed in as **display**	Opens `/display/<event>` in full screen (F11).

2. Accounts, logins and security

- There is **no sign-up**. Every account is created by the seed command on the organiser laptop.
- **Teams**: one shared login per team, printed on a card (team code + password). **Staff**: an email and a password, printed on a staff sheet (one slip each). Roles: **ORGANISER**, **FAIRNESS** (the faculty officer: reads everything, decides flags, approves corrections) and **DISPLAY** (read-only projector).
- Everyone signs in at the same address. Teams use the **Team** tab; staff use the **Organiser** tab.



After an organiser signs in: the list of events (a rehearsal is marked “Rehearsal ×10”).

The three rules that protect the event

1. Never paste a key, a password or the `.env` file into a chat, an email or GitHub.
2. `CARD_SECRET` (which can recreate every password) lives **only** on the organiser laptop, never on the hosting service.
3. The `out/` folder holds the printed passwords and the lottery seed. It is never committed, uploaded or shared. Back it up encrypted.

Printing the logins

Command (organiser laptop)	Writes
`pnpm seed --slug … --date … --teams … --staff … --commitment …`	The event, every login, the decks and the twist; `out/<slug>-login-cards.pdf` (8 cards per A4) and `out/<slug>-staff-logins.pdf` (one slip per staff member).
`pnpm cards … --only T007,T112`	Reprints chosen team cards (same passwords).

Command (organiser laptop)	Writes
`pnpm staff-sheet --slug …`	Reprints the staff sheet from the seed's text file.

3. Before the night

Two weeks before

- Accounts with two-factor sign-in and a second organiser on each: the database (Supabase), the website host and the Anthropic console (the AI judge). Fund the Anthropic account and ask for the rate limits you need.
- The database is set up and the website deployed (see docs/DEPLOY.md). A test sign-in works.

The week before: rehearse

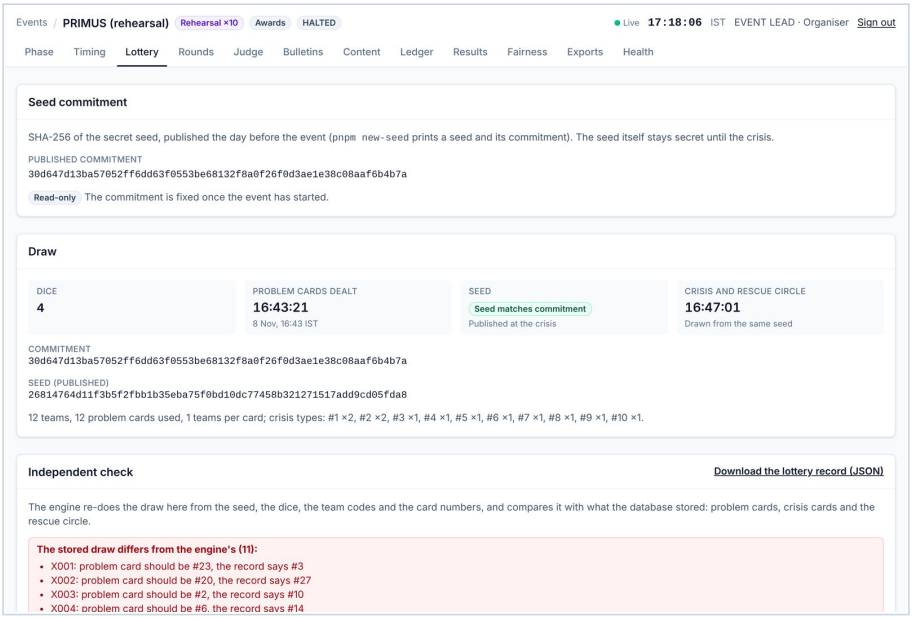
- Seed a rehearsal event (team codes start with X, every duration ÷10, so the night lasts about an hour) and run `pnpm rehearse --slug rehearsal --autopilot`: 150 bot teams play while you watch.
- Keep a few teams for people and play them on phones, with the projector open.
- Practise: starting the night, changing a duration, ending a phase by hand, pausing and resuming, closing a round early, a judging re-run, switching to the backup model, the fairness decisions, the awards and **End the night**.
- Run the load test (1,000 screens, a last-minute order burst): every clearing under 5 s, every screen updated within 2 s.

At least three days before: calibrate the judge

Judge → **Upload the samples** → **Judge them**, with **Run the judge here** on. Eight sample submissions, weak to excellent, are scored by each round's model, and the page shows whether each lands in its expected range. If several are out of range, tell the developer. Never change the ranges to make it pass.

The day before: the lottery seed

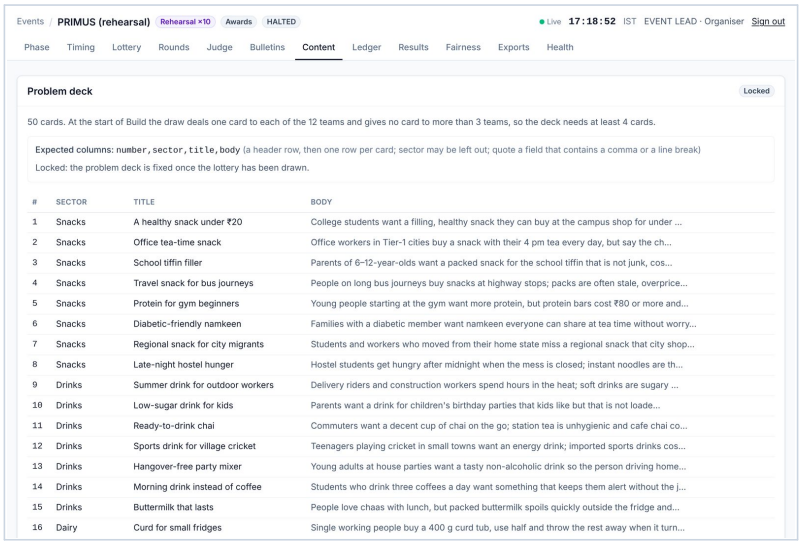
1. Run `pnpm new-seed`. It prints a **secret seed** and its **commitment** (a SHA-256 hash).
2. Give the seed to the event lead only: a sealed envelope or a password manager. It decides every draw.
3. Publish the commitment to every team (email, notice board), and enter it on the **Lottery** tab. Later anyone can check that the draws were not changed.



Lottery: the published commitment, the draw (seed + dice), the crisis draw and the public record. (After the rehearsal: the seed is public once the crisis has happened.)

Content

- **Problem deck**: at least 50 cards (sector, title, text). The draw gives no card to more than 3 teams.
- **Crisis deck**: 10 cards, one per crisis type; each type goes to the same number of companies.
- **The twist**: written before the night; editable on Content until it is published.
- Two **rumours** written and ready under Bulletins.



Content: the problem deck (locked once drawn), the crisis deck and the twist.

The evening checklist (by 19:30)

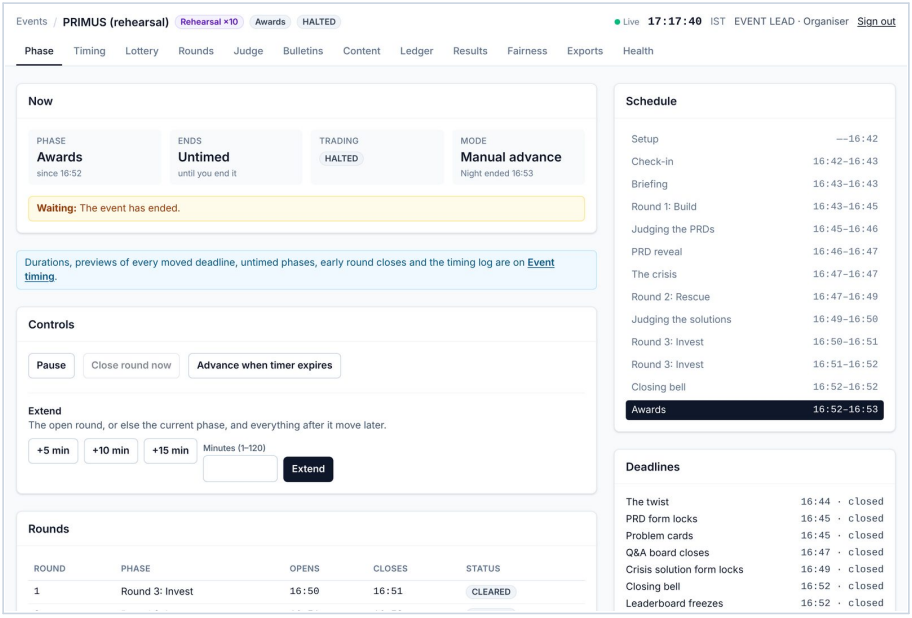
- **Health**: the heartbeat is green, no tick errors, the event is in Setup.
- **Lottery**: the commitment shows and matches the published one. The seed is with the event lead.
- **Phase**: no “Waiting” notice (the Setup gate is clear: commitment, decks and twist are in).
- **Judge**: a calibration batch from this week, samples in range; both rounds on the primary model; no FAKE JUDGE badge.

- **Timing**: the durations are right, and you have chosen **Manual advance** or **Advance when timer expires**.
- **Projector**: signed in as display, full screen, showing “Welcome to PRIMUS”.
- A test team card works on a phone on the venue Wi-Fi and on mobile data.

4. The control panel

Every organiser page has the same header: the event, its phase, the trading state, the live dot (green when the screen is connected), the server clock, and the tabs.

Tab	What it is for
Phase	Where the night is, what holds the next step, quick controls, rounds and the schedule.
Timing	**Event timing**: start and end the night, durations, previews, pause, extend, early round close, the timing log.
Lottery	The commitment, the draw (seed + dice) and the public lottery record.
Rounds	The nine trading rounds and every clearing (prices before and after, net shares).
Judge	The judge worker, each round's progress, model switch, retries, re-runs, appeals, calibration.
Bulletins	Publish rumours and notices; the twist; moderate the Q&A board.
Content	Problem deck, crisis deck, the twist.
Ledger	Every cash and share movement, with filters; corrections (one organiser requests, a second approves).
Results	Totals, ranks and awards (staff see them before the awards).
Fairness	Collusion flags and decisions (the fairness officer decides).
Exports	CSV downloads: ledger, prices, scores, results, awards, and the lottery record.
Health	Heartbeat, tick errors, the judge queue, connected screens.

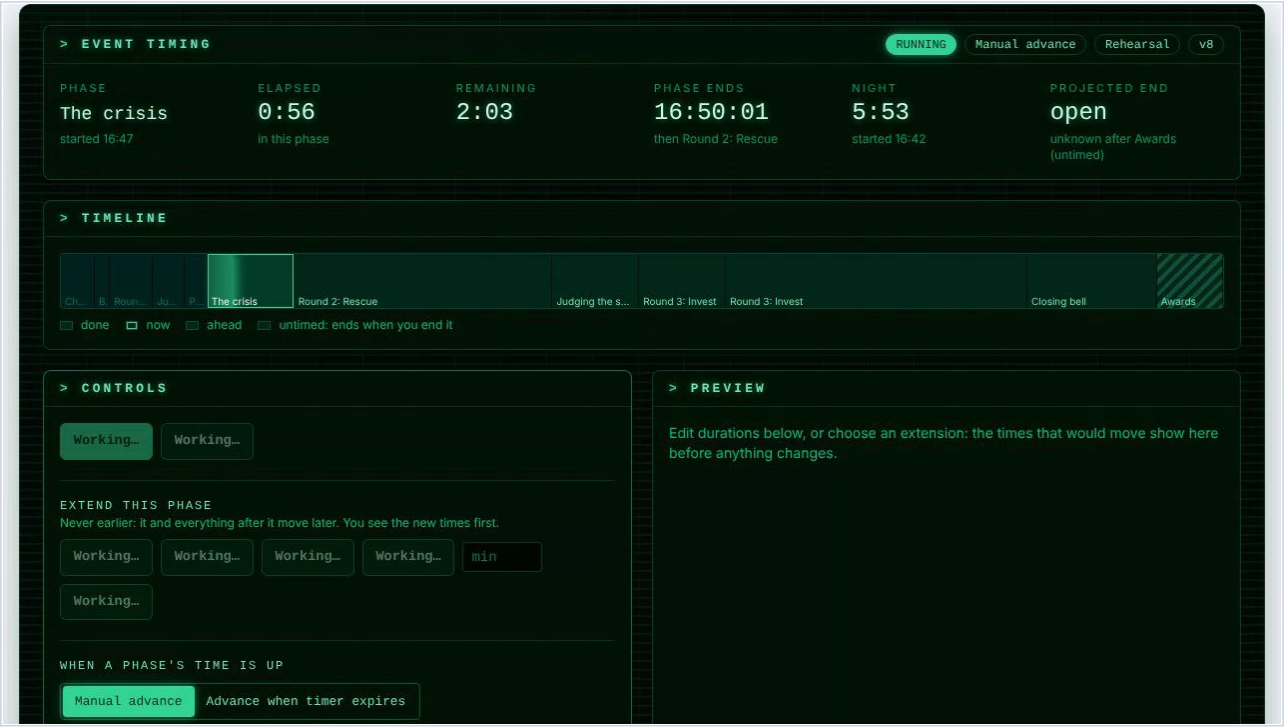


Phase: the night's status, controls and schedule (after the rehearsal ended).

> **TIP**
> Two-click buttons: anything that changes the night asks twice (“Click again to …”). The second click counts only for what the first one showed: if the page changes in between, it asks again.

5. Event timing

The **Timing** tab is the night's clock, in a dark console style. Everything on it comes from the database, which works out every time from what actually happened. The panel only edits durations and sends them back.



Event timing while the crisis is running: the live readout, the timeline (the current phase lit), controls and preview.

The readout

Reading	Meaning
Phase	The current phase and when it started.
Elapsed	How long the phase has run, not counting pauses.
Remaining	Time left (amber in the last minute). “untimed” for a phase you end by hand; “until the last round closes” in trading.
Phase ends	The projected end, and the phase after it.
Night	How long since you started the night.
Projected end	When the whole night ends, or the phase after which it is unknown (because a later phase is untimed).
Chips	RUNNING / PAUSED / ENDED, the mode, and the timing version (v12…), which changes with every change.

Starting the night

1. On Timing, click **Start the night…**. The panel shows a preview of every time that will follow, phase by phase and round by round. Anything still missing from Setup is shown as a

requirement.

2. Click **Start now**, then **Click again: start the night now**. Check-in starts at once on every screen.

Ending a phase

Click **End <phase>…**. Before anything happens you see: the next phase; any **requirement not met** (the lottery not drawn, scores not sealed, flags open); in a trading phase, the open round and how many pending orders will fill; and every time that moves. Confirm with two clicks.

Manual advance, or advance when the timer expires

- **Manual advance**: when a phase's time is up, the next phase waits for you (“time is up: waiting for you”).
- **Advance when timer expires**: the next phase starts by itself, unless something holds it.
- In both modes a timer that runs out still **locks its form or clears its round**. Manual only waits before starting the next phase. Switching to automatic when the time is already up asks for a second click, because it advances at once.

Changing durations

1. In **Phases and rounds**, type a new length in minutes (decimals like 1.5 are fine for rehearsals), or tick **Untimed** for a phase you will end by hand. Edited boxes glow amber; a wrong entry is marked on its row.
2. You can also change the **twist** (minutes after Build starts) and the **leaderboard freeze** (minutes after Market starts).
3. Click **Preview changes**. The Preview box lists every time that would move, from → to, with what teams will see marked “teams” and anything that comes **earlier** in amber.
4. Click **Apply**. If anything comes earlier, the button asks “Click again: bring N times earlier”. Teams are told immediately which of their deadlines moved.



The duration editor: every phase and round with its start, end and length in minutes (rehearsal lengths shown).

> **WATCH OUT**
> What cannot be changed: Setup (the night starts when you start it), the trading phases (they last as long as their rounds: change the rounds), a phase or round that has ended, and a length shorter than what has already run (end the phase instead). A twist or freeze that has happened is history.

Pause, resume and extend

- **Pause** (two clicks) freezes every countdown on every screen and refuses every save and order. **Resume** gives back exactly the time that was left. Tell the room before you resume.
- **Extend** (+1, +5, +10, +15 min or any number up to 120) lengthens the **open round**, or else the current phase. Everything after it moves later. You see the preview first. An untimed phase cannot be extended (it has no end).

Closing a round early

While a round is open, **Close a round early** shows how many orders are pending and offers two ways:

- **Close and clear**: every pending order fills at the round's clearing price, exactly as at its normal close.
- **Close and cancel**: every pending order is cancelled; nothing is bought or sold and the round clears as if none had been placed.

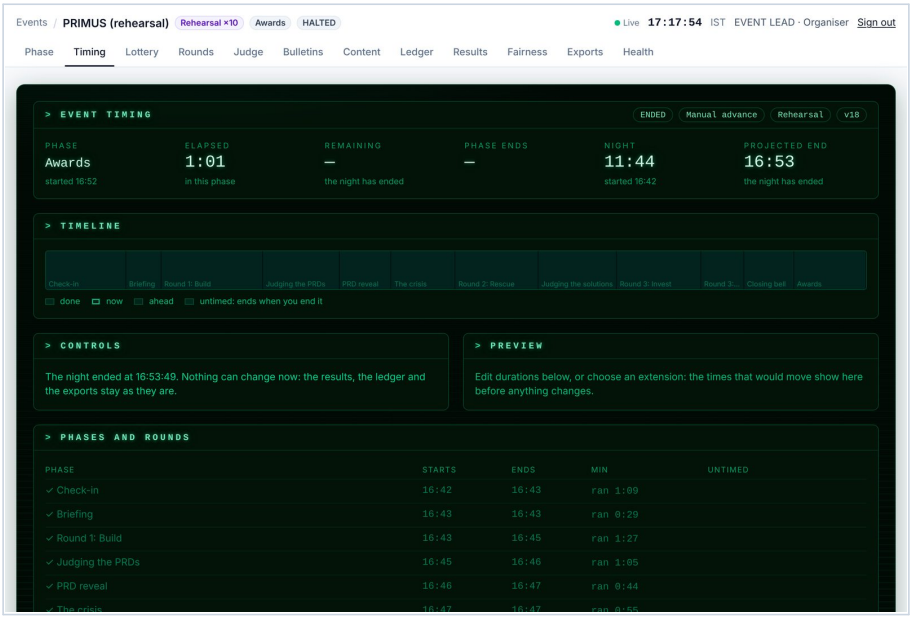
The rounds after it start from the actual close. The button names the round you saw: if that round has already closed (another console, the clock), nothing happens to the next one.

Two consoles at once

Every change carries the timing version your page showed. If another console changed the timing first, yours is refused (“The timing was changed on another console”) and your page shows the new times; look, then try again. Every change is in the **Timing log** with who did it and how many times moved.

Ending the night

In Awards (untimed), when the last award has been shown, click **End the night** twice. Every screen says the night is over, and nothing can change afterwards: results, ledger and exports stay as they are.



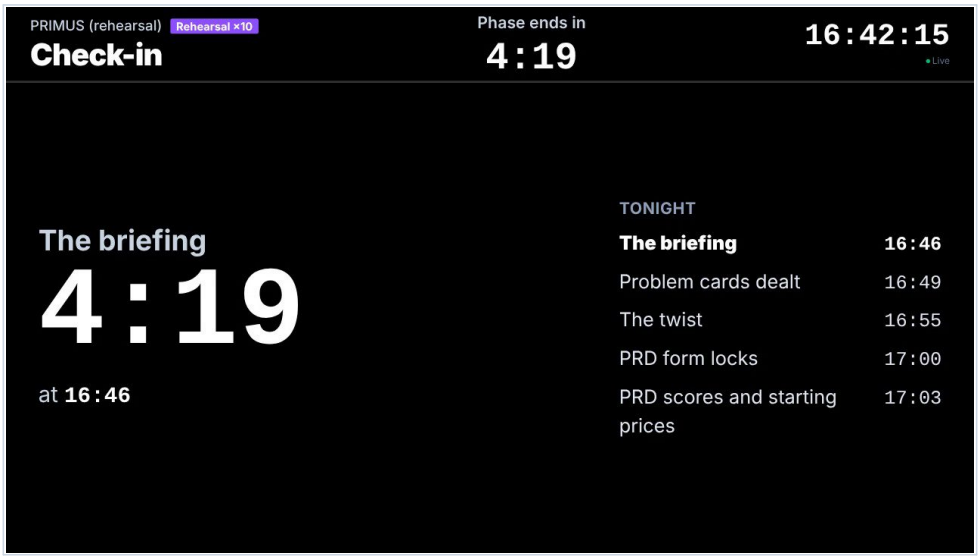
Event timing after the end of the night: ENDED, the elapsed time fixed, the log of every change.

6. Running the night, phase by phase

Each step says what **you** do and what the system does **by itself** (the database clock ticks every 2 seconds; an open organiser console ticks it too, as a backup). Keep the Timing tab open on laptop 1 all night.

Check-in (20:00)

You	Start the night (Timing). Hand out one card per team.
By itself	Check-in starts on every screen; teams sign in.
Watch	Health → connected screens rising towards 600.



The projector at check-in: the countdown and the evening's moments.

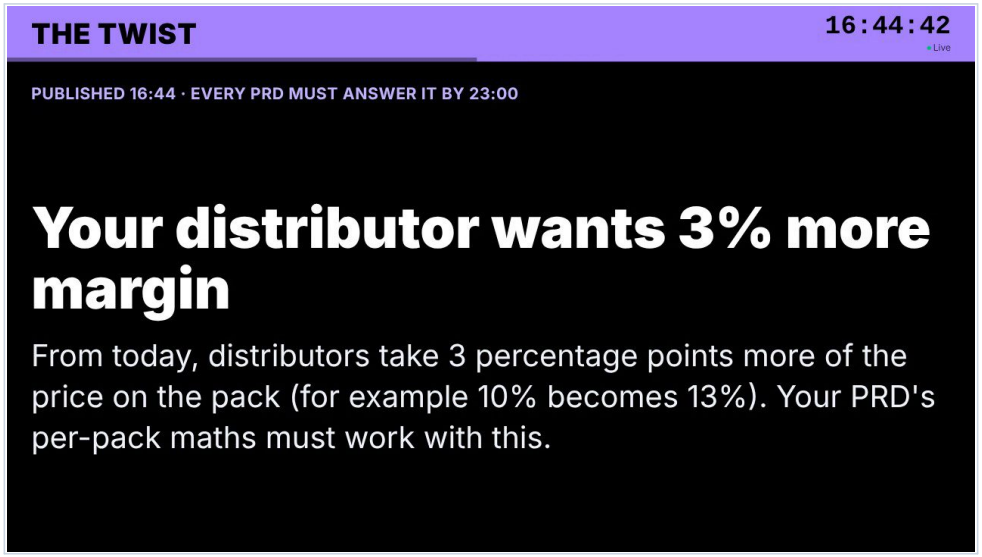
Briefing (20:45)

You	Explain the night; show “How judging works” (the team page) on the projector. Then, on stage: **Lottery** → enter the secret seed and a dice roll → **Verify the seed and draw** (two clicks).

By itself	The server checks the seed against the commitment, then deals the problem cards. Build cannot start before the draw.
Watch	Lottery: “Seed matches commitment”. Then end Briefing to start Build.

Round 1: Build (21:15)

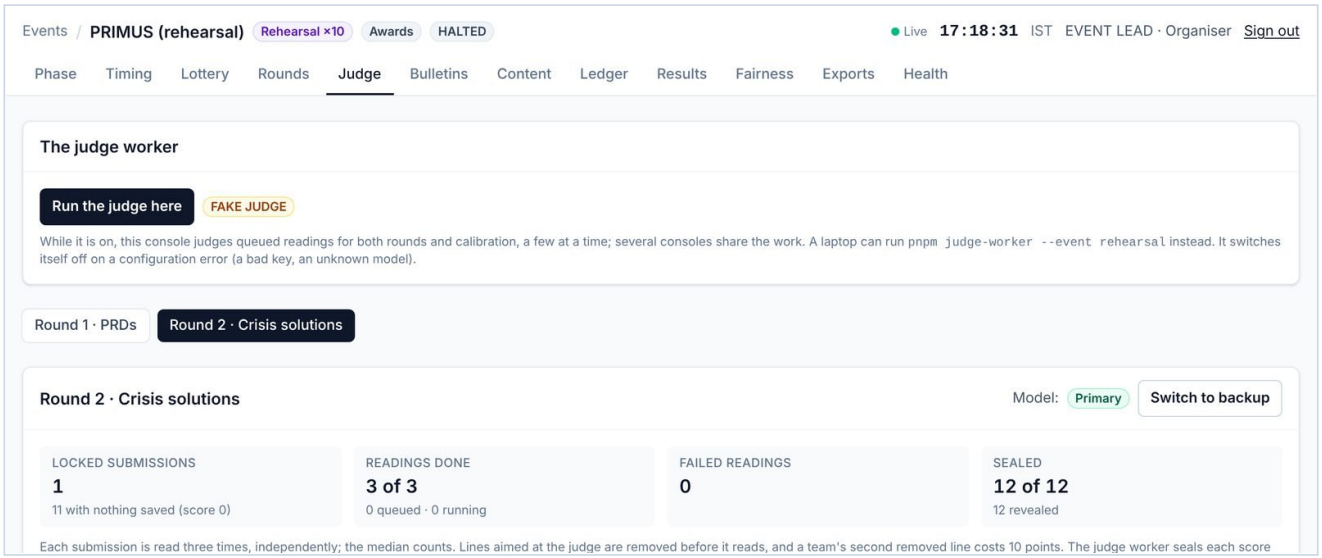
You	Nothing, unless you move the twist or the lock.
By itself	The twist is published to every screen 60 minutes in (a takeover). At the end the PRD form locks; three readings per PRD are queued.
Watch	The twist takeover on the projector.



The twist takeover on the projector.

Judging the PRDs (23:00)

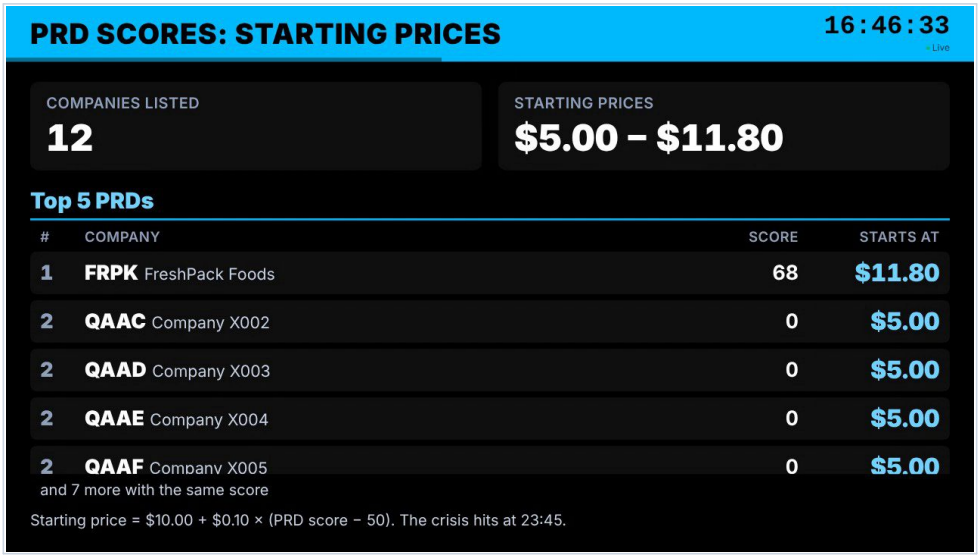
You	**Judge → Run the judge here** on laptops 2 and 3. Keep the tab open.
By itself	Each PRD is read three times; the median is sealed. Teams with nothing saved are sealed at 0.
Watch	Judge: readings done climbing; every score sealed in about 20 minutes. The reveal waits for the last one.



Judge: the worker (Run the judge here), each round's progress and every company's readings.

PRD reveal (23:30)

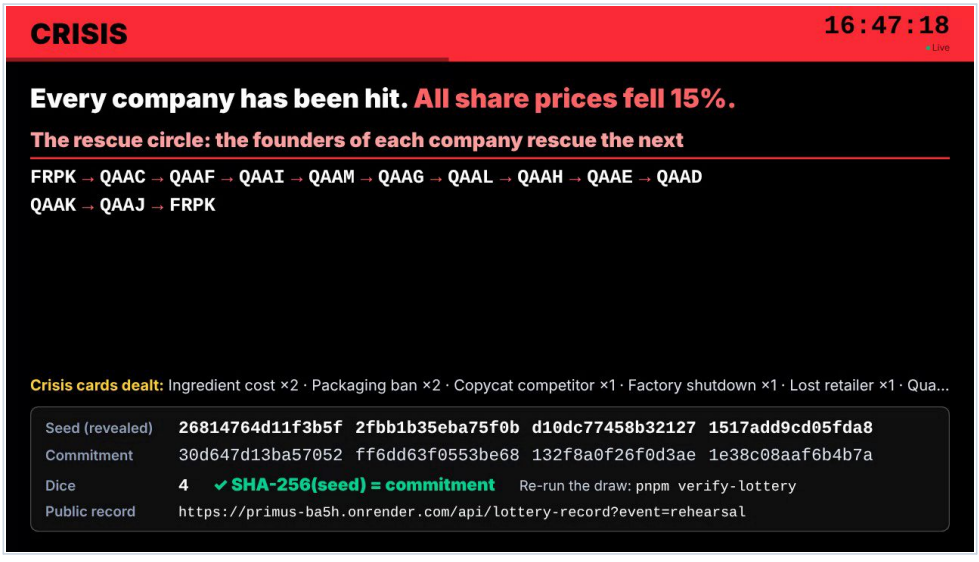
You	Nothing.
By itself	Starting prices and the top 5 PRDs on the projector; teams see their score and reason.
Watch	The reveal on the projector.



The PRD reveal: the range of starting prices and the top PRDs.

The crisis (23:45)

You	Nothing; announce it.
By itself	Crisis cards dealt, every price −15%, the rescue circle drawn, the seed published (the lottery record is public). The Q&A board opens.
Watch	The crisis takeover.

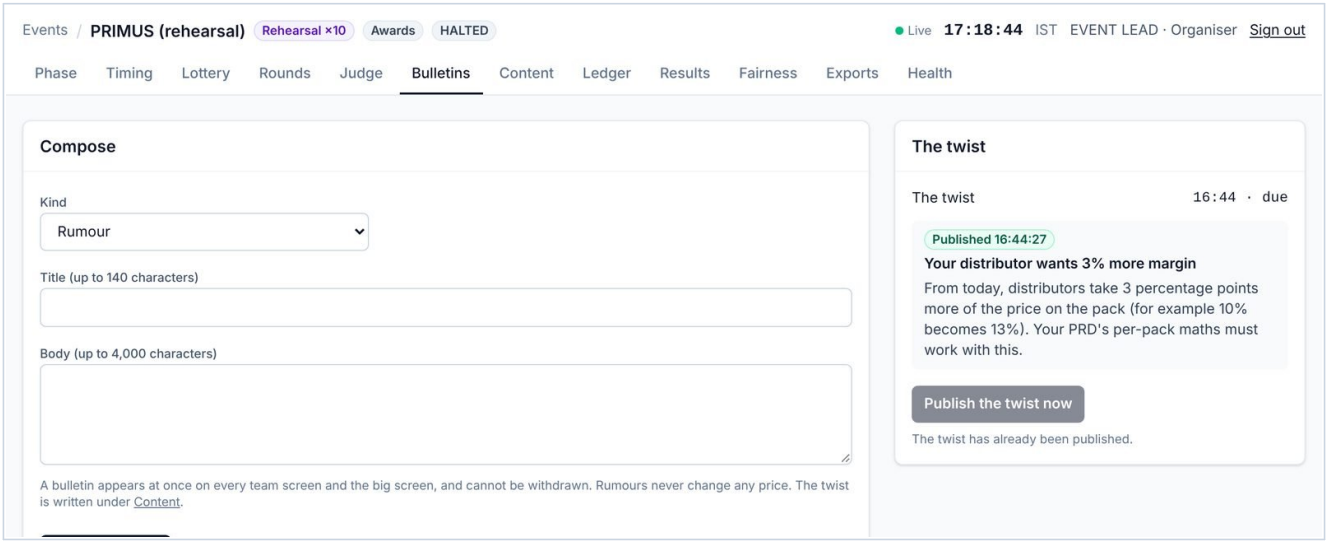


The crisis takeover, with the rescue circle and the published seed.

Round 2: Rescue (00:15)

You	Publish a rumour at about 00:30 and 01:00 (**Bulletins** → kind Rumour → **Publish bulletin**).
By itself	Questions close; solutions are written. At the end the solution form locks and readings are queued.

Watch	Keep the judge running from the lock.



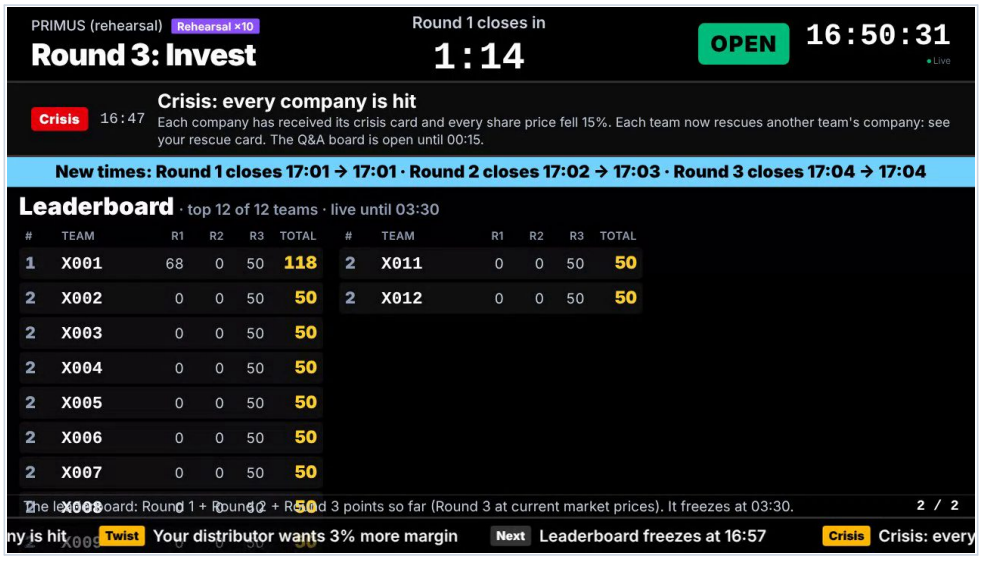
Bulletins: compose and publish a rumour; the twist; the Q&A board.

Judging the solutions (01:45)

You	**Run the judge here** on.
By itself	Readings sealed; scores stay secret.
Watch	Every score sealed well before the market's score reveal.

Invest, blind (02:15)

You	Nothing.
By itself	All solutions published without scores; trading rounds 1 and 2 open and clear back to back.
Watch	Rounds: each round clears within seconds of its close.

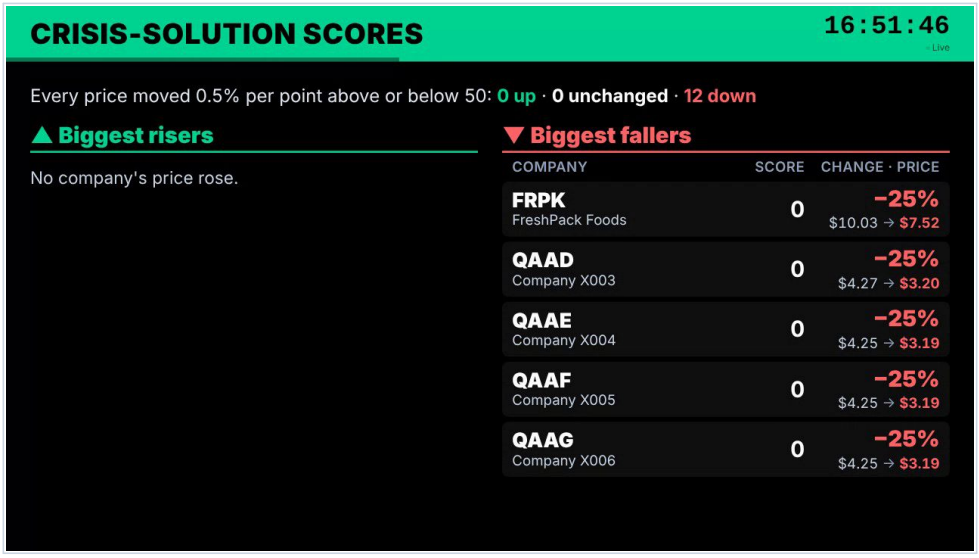


The price board during round 1, with the news ticker.

Invest (02:45)

You	Nothing (end the blind phase by hand in manual mode).

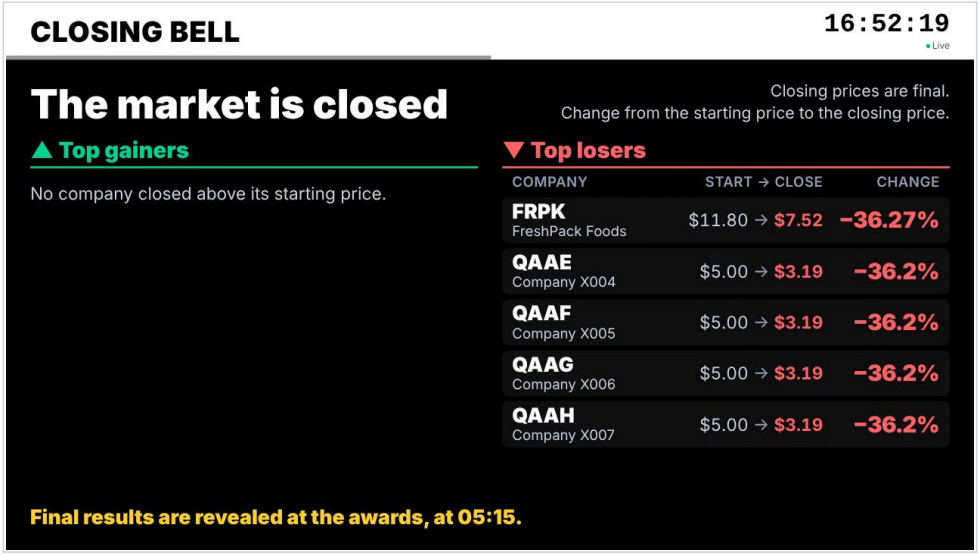
By itself	The score reveal moves every price; rounds 3–9; the leaderboard freezes 45 minutes in.
Watch	The score reveal takeover; the Rounds tab.



The score reveal: biggest risers and fallers.

Closing bell (04:30)

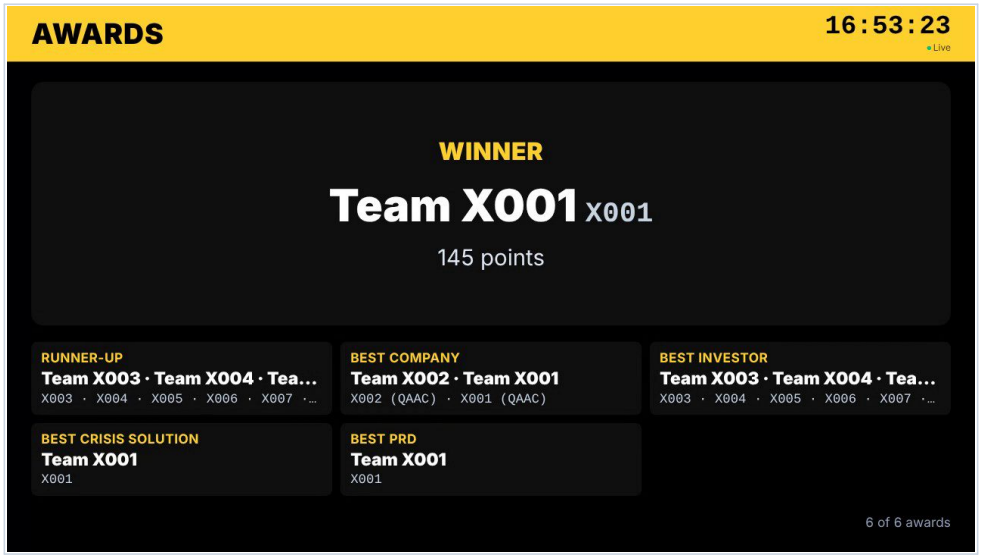
You	Fairness officer: review every flag (section 8).
By itself	Closing prices, Round 3 points, totals, awards, collusion flags.
Watch	Phase: what still holds the awards (open flags).



The closing bell on the projector.

Awards (05:15)

You	Let the awards play; then **End the night**. Afterwards: `pnpm certificates --event <slug>`.
By itself	The awards are revealed one by one on the projector, the winner last.
Watch	Every screen says the night is over.



The awards (rehearsal with one playing team).

7. The AI judge

- **Models**: PRDs by Claude Opus 5.5, crisis solutions by Claude Sonnet 5.5, each with a backup model.
- **Three readings** per submission, independent; the **median** counts. Each reading returns points per rubric line, a total and a three-sentence reason, in a fixed format checked by the server.
- **Anonymised**: team, member and company names are replaced by the ticker before the judge reads anything.
- **Context**: Round 1 gets the problem card and the twist; Round 2 gets the rescued company's PRD and crisis card.
- **No tricks**: lines aimed at the judge are removed and logged. A team's second attempt costs 10 points in that round.
- **Published**: the rubrics and exact instructions are on the teams' “How judging works” page.

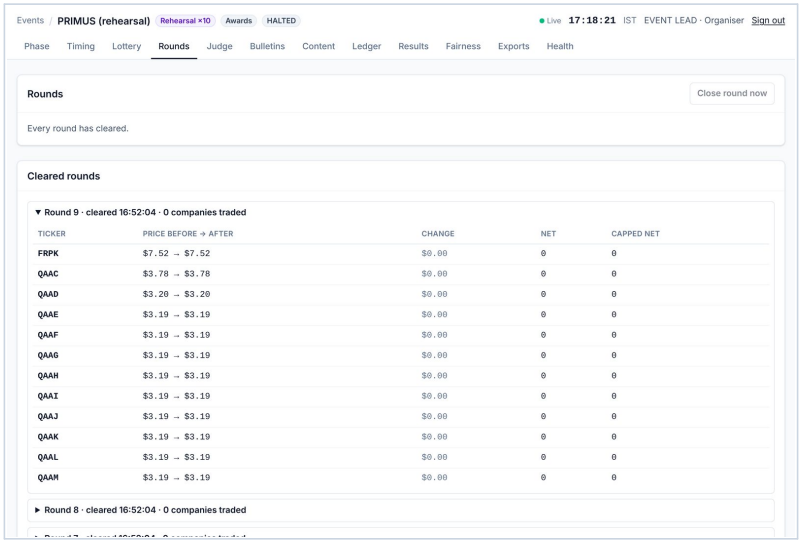
When something goes wrong with judging

Situation	Do
Judging is slow	Switch “Run the judge here” on a third laptop, or run `pnpm judge-worker --event <slug>`.
“The judge stopped: …” (configuration)	A bad API key or model name: fix the hosting variable, redeploy, switch the judge on again. The readings went back to the queue.
Many failed readings	Judge → **Switch to backup** for that round, then **Retry failed readings**.
The AI service is down	Extend the judging phase; switch the judge off. Three human judges score on paper against the rubric; enter the marks with `pnpm human-scores --event <slug> --type PRD --file prd.csv`. Scores seal and reveal as usual.
A submission must be re-judged	Judge → the company → **Re-run**, with the reason. Before its reveal the new score replaces the old; after it, a **second** organiser approves or dismisses it under “Re-runs and appeals”, and approving re-applies its price effect.

8. Market, ledger and fairness

How the market works (for answering questions)

- Starting price = $10.00 + $0.10 × (PRD score − 50). The crisis multiplies every price by 0.85.
- Score reveal: × (1 + 0.5% × (crisis score − 50)).
- Each round: net = shares bought − sold; the price moves (net ÷ 1,000)%, capped at ±10%; every order fills at the new price.
- Limits: no trading the company you built or rescue; no short selling; at most 5,000 shares of one company; a buy reserves price × 1.10.
- Closing price = ½ × (average of rounds 8 and 9) + ½ × score-based price. Round 3 points = 50 + 2.5 × return %, 0–100.



Rounds: every clearing, company by company (price before and after, net shares).

Ledger corrections

Nobody edits a price, a score, cash or shares directly. If a team's money or shares are wrong: **Ledger → Corrections**: one organiser requests the correction with a reason; a **second** approves it. Every step is in the audit log.

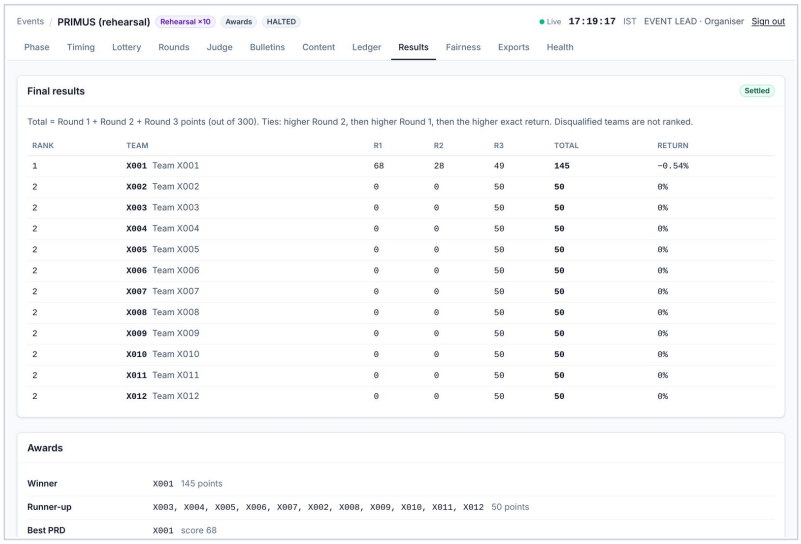
Collusion flags (fairness officer)

Flag	Raised when
1. Reciprocal buying	Two teams each bought the other's company.
2. Large losing position	A team holds 4,000+ shares of a company that closed below the team's average cost.
3. Crowd at the limit	Five or more teams reached the 5,000-share limit on the same company in the same round.

At the closing bell the fairness officer opens **Fairness**, reviews each flag (with the teams' full trade history), and decides **Cleared** or **Disqualified** with a written reason (two clicks). A disqualified team is removed from every ranking and award. The awards wait until every flag is decided.

9. Results, awards and after the night

- Total = Round 1 + Round 2 + Round 3 points (out of 300). Ties: higher Round 2, then higher Round 1, then higher exact return.
- Awards: winner, runner-up, third; Best PRD; Best crisis solution; Best investor; Best company (shared by its founders and rescuers).
- The leaderboard is live for teams until the freeze; the final ranking is hidden until the awards (staff see it on **Results**).



Results: the final totals and the awards as staff see them.

After the awards

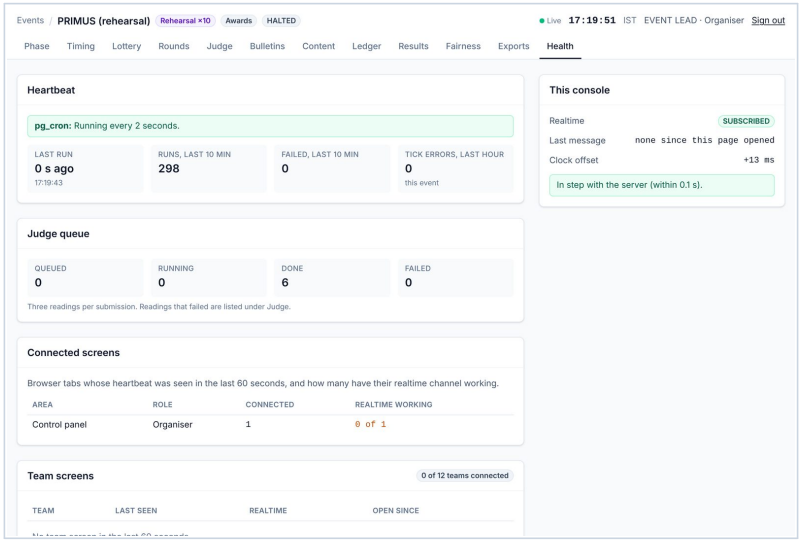
1. Print the certificates: `pnpm certificates --event <slug>` (one page per participant).
2. **Exports**: download the ledger, prices, scores, results and awards (CSV) and the public lottery record; store them encrypted.
3. Replace the AI and database secret keys; delete `out/` from the laptop once archived.
4. Purge rehearsal events and scale the hosting plans back down.

10. When something goes wrong

> **FIRST**
> **First rule: pause, then think.** Pause freezes every countdown and refuses every action; resume gives back exactly the time that was left.

Symptom	Likely cause	Do
A lock, reveal or round does not happen on time	The database clock stopped	Keep the Timing tab open (it ticks too). Health shows the heartbeat. A round past its close: Timing → Close a round early.
The phase does not move on	Something holds it, or manual mode	The End-phase preview says what: the lottery, unsealed scores, open flags. Or switch to Advance when timer expires.
Many teams cannot sign in	Venue Wi-Fi, or the sign-in rate limit	Mobile data; raise the auth rate limits; ask venue IT. Reprint a lost card.
Screens show a grey live dot	Realtime limit or the network	Screens re-read by themselves; nothing is lost. Reload a projector that stays grey.

Symptom	Likely cause	Do
The website is down	The hosting service	Pause (database SQL editor: `select public.pause_event('<event id>');`). Roll back the deployment; resume.
The database is down	A provider incident	Nothing changes while it is down. Announce, wait, then extend by the lost minutes.
The room needs more time	A phase is behind	Timing → Extend, or make the phase longer, or untimed and end it by hand.
A team's money or shares are wrong	A mistake	Ledger → Corrections: one requests, a second approves.
Power cut	Power	Pause. Move the projector and organiser desk to backup power; resume when teams can work.



Health: the heartbeat, tick errors, the judge queue and connected screens.